Three-stage analysis workflow

Stage 1 — Boundary-aware candidate sampling — §3.1

Per-city AOI = ADM2 prefecture polygon
6 cities · taken whole · no urban mask

ESA WorldCover 2021 → 3 classes
stratified sample 5,000 / class → 15,000 / city

Sentinel-2 2021 composite
9 spectral features (6 bands + NDVI, NDBI, MNDWI)
point value + 3×3 mean & SD

BAMS selection (two steps, not a sum)
1) keep 1,000 lowest-margin ($m = p_1 - p_2$)
2) of those, top 150 by BoundaryScore = $\Sigma\, \sigma_{3\times3}$(B2, B3, B4, B8)

the random forest is a proposal mechanism only — it assigns no class labels

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900 boundary points · independent expert photo-interpretation

QC: 120-pt re-label, Fleiss' κ = 0.89 (116 complete cases), 92.7% mean pairwise

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Stage 2 — Multi-reference agreement diagnostic — core — §3.2

Does consulting a 2nd / 3rd product (Dynamic World, Esri) arbitrate truth at the urban boundary?

compare expert vs WorldCover · Dynamic World · Esri (n = 900)

(a) Four-way agreement
"how much"
Table 2

(b) Confidence stratification
annotator noise?
Table 3

(c) Direction of WC error
"which way"
Table 4

(d) Physical scene stratification
"where"
Table 5

(e) Independence & arbitration test
excess agreement · error dependence (Yule Q, κ) · arbitration value · confounds
Table 6 · Fig 3 · Tables S3–S4

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Stage 3 — LOCO correction operator — supporting — §3.3

leave-one-city-out: fit expert corrections on 5 cities → apply to the held-out 6th (no labels of its own)

the operator overwrites the WorldCover label only where it is confident (gate τ)

evaluation protocol: same fixed downstream classifier throughout · spatial-block train/test splits · target-city isolation

Phase 1
operator transfer

Phase 2
downstream classifier training

Phase 3
multi-reference referee

before / after correction maps: Fig 4 (main) · Fig S1–S4

All agreement / disagreement rates are conditional on the boundary-candidate population; they are not map-wide accuracies of any product.